

AI Internship Report on
Development of an FAQ Chatbot and a Real-Time Object Detection
& Tracking System

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By

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CANDIDATE'S DECLARATION

I, Satwik Kumar, do hereby declare that the internship report titled “Development of an FAQ Chatbot and a Real-Time Object Detection & Tracking System”, based on the Artificial Intelligence Virtual Internship undertaken at CodeAlpha (Student ID: CA/DF1/186503) from 10th July 2026 to 10th August 2026, has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own and has not been submitted to any other institute for any degree/diploma requirement, and I shall be solely responsible for any kind of copyright violation in this regard.

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ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program at CodeAlpha. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to my mentors at CodeAlpha for sharing their truthful and illuminating views on a number of issues related to this project.

I would also like to thank the School of Computer Science and Engineering, IILM University, for providing me the opportunity to undertake this internship as part of my academic curriculum, and for the continuous encouragement extended by the faculty.

I express my gratitude to my parents for their blessings and constant support.

Signature: _____

Student Name: Satwik Kumar

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INTERNSHIP COMPLETION CERTIFICATE

 YOUR DREAM OUR PASSION	 Student ID: CA/DF1/186503
CERTIFICATE OF COMPLETION	
This Certificate Is Proudly Presented To <i>Satwik Kumar</i>	
Was an active Participant at CodeAlpha Virtual Internship Program in Artificial Intelligence Internship with dedication and hard work. We really appreciate your efforts taken and wish you all the best for the further. 10th July 2026 to 10th August 2026	
 CEO & Founder	29th August 2026 Date of Issue 

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4. Project Description

4.1 Introduction

This report presents the work carried out during my Artificial Intelligence internship at CodeAlpha, a software development company focused on driving innovation across emerging technologies such as artificial intelligence, machine learning, and real-time data processing. As part of the internship program, I completed two of the four assigned tasks from the AI domain: Task 2 – Chatbot for FAQs, and Task 4 – Object Detection and Tracking. Both projects were undertaken independently, following the task instructions provided by CodeAlpha, and involved end-to-end development — from problem understanding and data preparation to implementation, testing, and documentation.

The internship was structured as a self-paced, project-based learning program in which interns build working AI applications, host the source code on GitHub, and submit a demonstration for evaluation. This report documents the objectives, methodology, technologies, architecture, and outcomes of the two completed projects.

4.2 Organization Profile

CodeAlpha is a software development company that offers virtual internship programs across domains such as Artificial Intelligence, Web Development, Android Development, Python Programming, Data Science, and Cybersecurity. The organization is dedicated to driving innovation and excellence across emerging technologies and aims to provide students with real-world, hands-on project experience rather than purely theoretical learning.

The AI track of the internship focuses on giving interns practical exposure to AI model development, machine learning workflows, and real-time data processing under expert mentorship. On successful completion of the required number of tasks, CodeAlpha issues an Offer Letter, a QR-verified Completion Certificate, a Unique ID Certificate, and, based on performance, a Letter of Recommendation, along with resume-building and placement support.

4.3 Problem Statement

Two independent problem areas were addressed as part of this internship:

- **Chatbot for FAQs:** Manually searching through lengthy FAQ documents to find an answer to a specific question is slow and inconvenient for users. There was a need for an automated system that could understand a user's natural-language question and instantly return the most relevant answer from a pre-defined FAQ knowledge base.
- **Object Detection and Tracking:** Manually monitoring a video feed to identify and follow objects of interest is impractical for real-time applications such as surveillance, traffic monitoring, or automation. There was a need for a system that could automatically detect multiple objects in a live video stream, classify them, and track their movement across frames with a consistent identity.

4.4 Project Objectives

- To design and build a text-based chatbot capable of answering user queries by matching them against a curated set of frequently asked questions.
- To apply natural language processing techniques for text cleaning, tokenization, and similarity matching between user input and stored FAQs.
- To build a real-time object detection pipeline using a pre-trained deep learning model capable of identifying multiple object classes from a live or recorded video feed.
- To integrate a multi-object tracking algorithm so that detected objects retain a consistent identity (tracking ID) across consecutive frames.
- To evaluate both systems for accuracy, responsiveness, and usability, and to document the complete development process.

4.5 Scope of the Project

The FAQ Chatbot is scoped as a domain-specific, retrieval-based question-answering system; it selects the best-matching answer from a fixed FAQ set rather than generating free-form answers, and its accuracy depends on the coverage of the underlying FAQ dataset. The Object Detection and Tracking system is scoped for identifying common, pre-trained object classes in real time from webcam or recorded video input; it is intended as a functional prototype demonstrating the detection-and-tracking pipeline rather than a production-grade surveillance system. Both projects were developed,

tested, and documented individually, with source code hosted on GitHub as per the internship submission guidelines.

4.6 Technologies and Tools Used

- Programming Language: Python 3
- NLP Libraries: NLTK / spaCy, scikit-learn (for TF-IDF and cosine similarity)
- Chatbot Interface: Streamlit / Flask (simple web-based chat UI)
- Computer Vision Library: OpenCV
- Object Detection Model: Pre-trained YOLO (You Only Look Once) model
- Object Tracking Algorithm: SORT (Simple Online and Realtime Tracking) / Deep SORT
- Development Tools: Jupyter Notebook / VS Code, Git and GitHub
- Version Control & Submission: GitHub repositories named as per CodeAlpha's naming convention (CodeAlpha_ProjectName)

4.7 System Architecture

4.7.1 FAQ Chatbot Architecture

The chatbot follows a simple retrieval-based pipeline: (1) a curated FAQ dataset of question–answer pairs is stored and pre-processed; (2) user input is captured through a text interface; (3) both the stored FAQ questions and the user's query are cleaned and vectorized (e.g., using TF-IDF); (4) cosine similarity is computed between the user's query vector and each FAQ question vector; (5) the FAQ with the highest similarity score above a set threshold is selected, and its corresponding answer is returned as the chatbot's response; if no match crosses the threshold, a fallback message is shown.

4.7.2 Object Detection and Tracking Architecture

The object detection and tracking pipeline works as follows: (1) video frames are captured in real time from a webcam or video file using OpenCV; (2) each frame is passed through a pre-trained YOLO model, which outputs bounding boxes, class labels, and confidence scores for detected objects; (3) low-confidence detections are filtered out; (4) the remaining detections are passed to a tracking algorithm (SORT/Deep SORT), which associates detections across frames using motion prediction and assigns

a persistent tracking ID to each object; (5) the annotated frame — with bounding boxes, class labels, and tracking IDs — is displayed to the user in real time.

4.8 Methodology

4.8.1 Project 1: Chatbot for FAQs

- Collected a set of frequently asked questions and their corresponding answers related to the chosen topic/product.
- Pre-processed the text data using an NLP library (NLTK/spaCy) — lower-casing, removing stop words and punctuation, and tokenizing the text.
- Converted the cleaned FAQ questions and user queries into numerical vectors using TF-IDF vectorization.
- Computed cosine similarity between the user's query vector and all FAQ question vectors to identify the closest match.
- Returned the answer corresponding to the best-matching FAQ as the chatbot's response, with a simple interface for interaction.
- Tested the chatbot with a variety of paraphrased and out-of-scope questions to evaluate matching accuracy and fallback behaviour.

4.8.2 Project 2: Object Detection and Tracking

- Set up real-time video capture using OpenCV, sourcing input from a webcam or a pre-recorded video file.
- Loaded a pre-trained YOLO model for detecting multiple common object classes in each frame.
- Processed every video frame through the model to generate bounding boxes, class labels, and confidence scores.
- Applied a tracking algorithm (SORT/Deep SORT) to maintain consistent object identities across frames as objects moved.
- Rendered bounding boxes, class labels, and tracking IDs on the live video output for visual verification.
- Tested the system with different object classes, lighting conditions, and movement speeds to evaluate detection and tracking robustness.

4.9 Expected Outcomes

- A working FAQ chatbot capable of accurately answering the majority of queries that closely match the underlying FAQ dataset, with a clear fallback response for unmatched queries.
- A functional real-time object detection and tracking system capable of identifying multiple object classes and maintaining consistent tracking IDs across video frames.
- Improved practical understanding of NLP techniques (text vectorization, similarity matching) and computer vision techniques (object detection, multi-object tracking).
- Two GitHub repositories (CodeAlpha_FAQChatbot and CodeAlpha_ObjectDetectionTracking) containing complete, documented source code as per CodeAlpha's submission guidelines.
- Fulfilment of CodeAlpha's internship completion criteria by successfully completing two of the four assigned AI tasks.

4.10 Certificates of Completion and Other Communication Mail Proof

The CodeAlpha Internship Completion Certificate (Student ID: CA/DF1/186503), issued for the period 10th July 2026 to 10th August 2026 and dated 29th August 2026, is attached in Chapter 3 of this report. Any additional communication/email proof from CodeAlpha may be appended after this section if required by the department.

5. Bibliography/References

CodeAlpha, “Artificial Intelligence Tasks & Instructions,” CodeAlpha official task document, www.codealpha.tech.

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